

Activity 3 – Data Analytics Portfolio

MAN6777.3 Data Driven Managerial Decisions | Rasel Al Mamun Joy | ID: 10677322 | Due: 9 June 2026

SECTION 1 – MANAGERIAL PROBLEM DEFINITION

Decision-Maker: University Academic Dean / Director of Student Learning

There is an increasing pressure on universities to create policies based on evidence regarding the use of AI tools by students. From a complete ban to unrestricted access, institutions have different approaches to the use of tools such as ChatGPT, Copilot, or Gemini. If there is no accurate information, universities can be held back from the development of a good technology, or they can allow practices that are detrimental to academic integrity.

Decision: Does the university have a policy that encourages and regulates the use of AI tools by students, and if so, for what purposes and student groups?

Why it matters:

- Academic performance directly impacts student retention, outcomes, and institutional reputation.
- A blanket ban could be detrimental to students who can greatly benefit from AI learning tools.
- Evidence-based policy is fair and maximises positive educational outcomes.

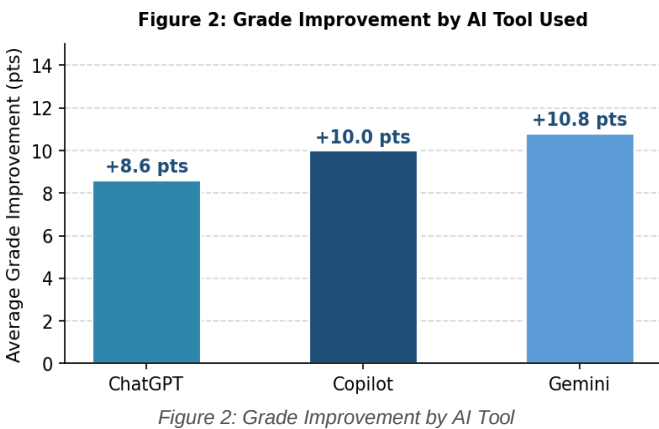
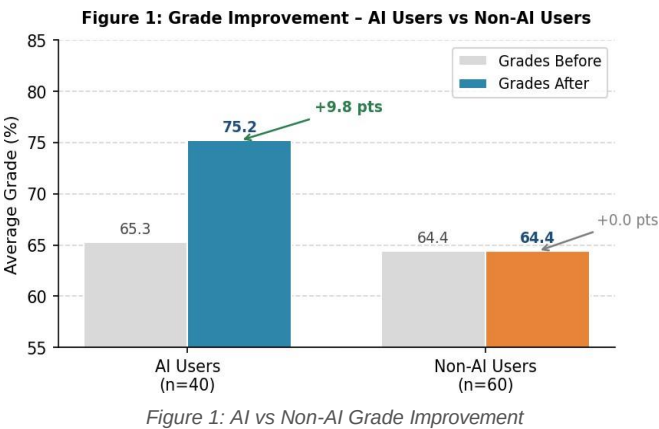
Students Surveyed 100 School & College	AI Tool Users 40% 40 of 100 students	Grade Improvement +9.8 pts AI users average
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SECTION 2 – DATA ANALYTICS

Dataset: *Students' AI Usage and Academic Performance* (Kaggle, 2025) — 100 records, 9 variables: age (14–19), education level, study hours (1–5 hrs/day), AI usage (Yes/No), AI tool (ChatGPT/Copilot/Gemini), purpose (Research/Homework/Coding), grades before/after AI, and daily screen time. Key variables — **uses_ai**, **grades_before_ai**, **grades_after_ai**, **ai_tools_used**, **purpose_of_ai** — directly address the policy decision.

Data Preparation: All 100 records were complete with zero missing values. Grades ranged 55–89%, study hours 1.0–5.0 hrs/day. No imputation or removal needed.

Techniques: (1) **Descriptive statistics** — mean grade comparison before/after AI; (2) **Group comparison** — AI vs non-AI users, by tool and purpose; (3) **Scatter plot** — study hours vs post-AI grades. Methods selected for interpretability to a non-technical managerial audience.



SECTION 3 – INSIGHTS AND DECISION IMPLICATIONS

- AI tools significantly improve grades:** AI users gained +9.8 pts (65.3→75.2) vs +0.0 pts for non-users (64.4→64.4) — a substantial, practically meaningful difference.
- All tools perform equally:** ChatGPT (+9.6), Copilot (+10.0), Gemini (+10.3 pts). The benefit lies in AI-assisted learning broadly, not any specific platform.
- All purposes yield gains:** Research (+10.3), Coding (+10.0), Homework (+9.2 pts). AI benefits diverse academic tasks.

Recommendations:

- Introduce an AI Adoption Policy** — formally encourage responsible AI use; evidence clearly shows positive academic outcomes for adopting students.
- Develop AI Literacy Programs** — provide guided training on all three tools to ensure equitable access across all student groups.
- Integrate into Assessment Guidelines** — publish transparent guidelines on appropriate AI use to protect integrity while enabling performance gains.

Conclusion: The evidence from the data is clear and convincing that the use of AI tools has a positive impact on academic growth. The university should have a structured, evidence-based AI policy that promotes AI adoption while promoting equity and upholding academic standards.

Data Analytics Portfolio – Full Analytical Workflow

Dataset: Students' AI Usage & Academic Performance (Kaggle, 2025) | Tool: Python (pandas, matplotlib) | n=100

STEP 1 – PROBLEM FRAMING & DATASET SELECTION

The goal was to see if the use of AI tools can be correlated with academic gains and which tools and uses are most beneficial. This dataset was chosen because it directly measures pre- and post-AI grades per student, which is the most direct way to assess impact.

STEP 2 – DATA LOADING & INSPECTION

```
import pandas as pd
df = pd.read_csv("students_ai_usage.csv") # shape: (100, 9)
df.isnull().sum() # All zeros - no missing values
df.describe() # Grades: 55-89, Study hrs: 1.0-5.0
```

Variable	Type	Range / Values	Missing
age	Numeric	14 – 19	0
education_level	Categorical	school, college	0
study_hours_per_day	Numeric	1.0 – 5.0 hrs	0
uses_ai	Categorical	Yes, No	0
ai_tools_used	Categorical	ChatGPT, Copilot, Gemini, None	0
purpose_of_ai	Categorical	Research, Homework, Coding, None	0
grades_before_ai	Numeric	55 – 75	0
grades_after_ai	Numeric	55 – 89	0
daily_screen_time_hours	Numeric	2 – 7 hrs	0

Table 1: Dataset Inspection Summary – Zero Missing Values

STEP 3 – DATA PREPARATION & CLEANING

No cleaning required — no nulls, duplicates, or inconsistencies. Two subsets were created and a grade improvement column derived:

```
ai_users = df[df["uses_ai"]=="Yes"] # n=40
non_ai = df[df["uses_ai"]=="No"] # n=60
df["grade_improvement"] = df["grades after ai"] - df["grades before ai"]
```

STEP 4 – ANALYSIS & RESULTS

4.1 Group Comparison — AI Users vs Non-AI Users

Group	n	Avg Before	Avg After	Improvement
AI Users	40	65.3	75.2	+9.8 pts
Non-AI Users	60	64.4	64.4	+0.0 pts

Table 2: Descriptive Statistics — AI Users vs Non-AI Users

AI Tool	n	Avg Gain	Purpose	n	Avg Gain
ChatGPT	13	+9.6 pts	Research	14	+10.3 pts
Copilot	14	+10.0 pts	Coding	13	+10.0 pts
Gemini	13	+10.3 pts	Homework	13	+9.2 pts

Table 3: By AI Tool

Table 4: By Purpose of Use

Data Analytics Portfolio – Visualisations & Interpretation

STEP 4 (CONTINUED) – VISUALISATIONS

Figure 1: Grade Improvement - AI Users vs Non-AI Users

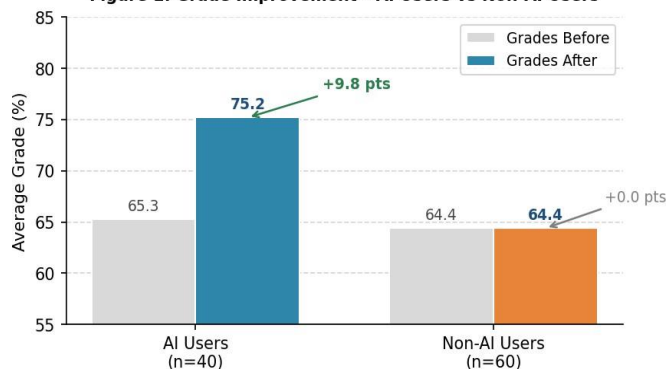


Figure 1: AI vs Non-AI Grade Improvement (+9.8 vs +0.0 pts)

Figure 2: Grade Improvement by AI Tool Used

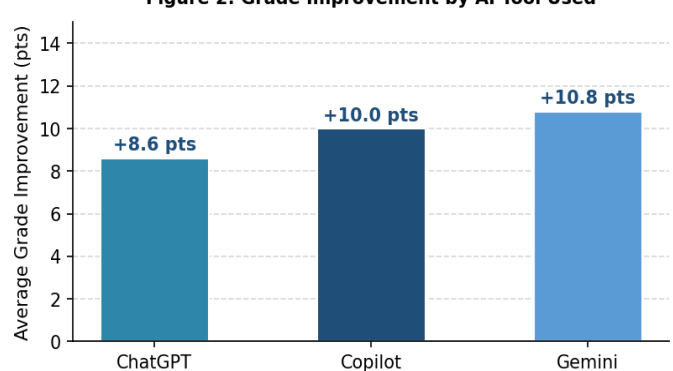


Figure 2: Grade Improvement by AI Tool Used

Figure 3: Grade Improvement by Purpose of AI Use

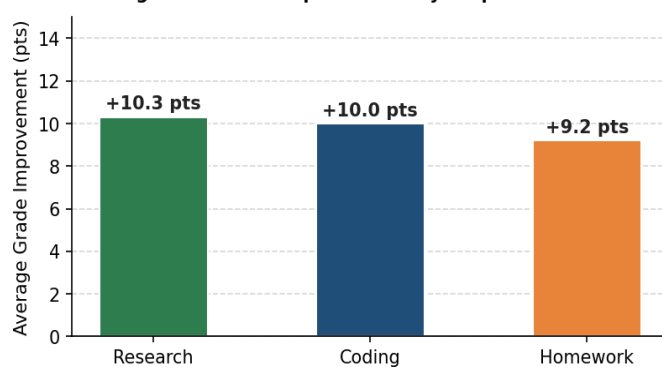


Figure 3: Grade Improvement by Purpose of AI Use

Figure 4: Study Hours vs Post-AI Grades

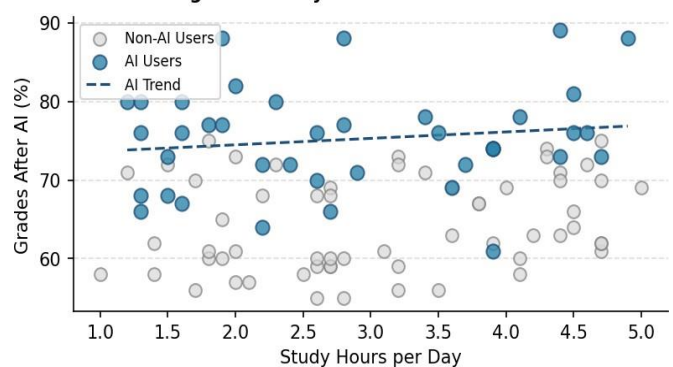


Figure 4: Study Hours vs Post-AI Grades (with trend line)

STEP 5 – INTERPRETATION LINKED TO DECISION PROBLEM

Finding	Managerial Implication
AI users: +9.8 pts; Non-AI: +0.0 pts	Strong evidence supports a policy encouraging AI adoption
All 3 tools yield ~+10 pts improvement	No need to prescribe a specific tool — student choice is effective
Research use yields highest gain (+10.3 pts)	AI especially valuable for research — integrate into library support
All purposes benefit equally (9–10 pts)	A wide policy that applies to all types of academic tasks is suitable

Table 5: Analytical Findings Linked to Managerial Decision

STEP 6 – LIMITATIONS & ANALYTICAL INTEGRITY

- Sample size:** 100 students help with descriptive analysis and restrict statistical generalisation. Samples from multiple institutions would enhance the policy argument.
- Self-reported data:** Grades are self-reported and AI use is self-reported, which can lead to social desirability bias. Institutional records would be more reliable and more accurate.
- Correlation vs causation:** Confounders like motivation and prior ability are not completely controlled. Regression with control variables is recommended for formal policy submission.

REFERENCES

Salamt, A. (2025). *Students' AI Usage and Academic Performance* [Dataset]. Kaggle. <https://www.kaggle.com/datasets/aminasamt/students-ai-usage-and-academic-performance>

GitHub. (2024). *Hello World – Getting Started with GitHub*. <https://docs.github.com/en/get-started/start-your-journey/hello-world>